THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME, AND AN INCOMPLETENESS IN THE STANDARD RECIPE FOR SCHOFER'S κ

Abstract. Schofer's formula expresses averages of $\log\|\Psi_F\|$ over CM points on a Shimura curve in terms of quantities $\kappa_\eta(m)$ built from derivatives of Fourier coefficients of an incoherent Eisenstein series. The recipe used in practice to evaluate $\kappa_\eta(m)$ descends from Kudla–Rapoport–Yang, through Schofer and Errthum, to Yang; its central step enumerates the configurations that contribute to the derivative at the centre. We observe that this enumeration is derived under a restriction which Kudla–Rapoport–Yang state explicitly — that the Eisenstein series be built from sections given by the characteristic functions of the completions of $\mathcal{O}_k$ — and which is not satisfied by the lattices arising for an Eichler order of squarefree level N at a *split* level prime.

We compute the local Whittaker factor of the relevant N-scaled binary lattice at the level prime in closed form for every valuation, and show that its value at the critical point is $(N-1)\operatorname{ord}_N(m)$. We then re-run the case analysis with this factor in place and find that it produces no additional contributing configuration, so that no $\log N$ can arise from $\kappa_\eta(m)$ for $m > 0$. Independently, a numerical evaluation of the vector-valued input form determines, exactly, a correction of the form $A_m \log N$ that the CM values demonstrably require. The rule producing it — half the constant term of the input form at a nonzero isotropic coset — is verified against ground truth on four bases with $N \in \{2, 3, 5\}$, including a base with $N = 5$ where the ground truth is forced by an over-determined consistency relation measured independently of any Eisenstein computation. We identify the correction as the constant-term pairing of the input form with the Eisenstein series attached to a nonzero isotropic coset of the discriminant form — an object the maximal-order literature does not compute — and reduce it, by an oldform relation of Schwagenscheidt, to ordinary seed-0 data of an enlarged lattice whose level-prime part is unimodular; this explains structurally both the support rule and the observed normalisation. We record the exact data any corrected formula must reproduce, and the routes we have eliminated. Finally, a finite exact evaluation of the constant term — expansions of the input form at all cusps of $\Gamma_0(M)$ against exact Weil-representation data, with no Fourier sampling — extends the measurements to bases beyond the numerical oracle's reach and shows that the correction carries a *second channel*, supported on the perfect-square indices of the principal part, invisible on every base where the level-support rule had been tested. The space of weight-1/2 forms for the Weil representation of the discriminant form is shown to vanish, so this channel is scalar — Zagier's square-class structure entering through the cusps — and not a harmonic completion on the discriminant form.

1. Introduction

Let B be an indefinite quaternion algebra over $\mathbb{Q}$ of reduced discriminant D, let $\mathcal{O} \subset B$ be an Eichler order of squarefree level N with $(N, D) = 1$, and let $L = \{\alpha \in \mathcal{O} : \operatorname{tr}\alpha = 0\}$ be the associated lattice of signature $(1, 2)$, so that $L^\vee/L \cong \mathbb{Z}/2 \times \mathbb{Z}/DN \times \mathbb{Z}/DN$ has order $2(DN)^2$ [4, Lemma 8]. For a weakly holomorphic vector-valued modular form F of weight $1/2$ and type ρ_L, Borcherds' lift Ψ_F is a meromorphic modular form on $X_0^D(N)/W_{D,N}$, and Schofer's formula [8, 4] evaluates

$$\sum_{\tau \in \mathrm{CM}(d)} \log|\psi_F(\tau)| \;=\; -\frac{|\mathrm{CM}(d)|}{4} \sum_{\eta \in L^\vee/L} \sum_{m \geq 0} c_\eta(-m)\,\kappa_\eta(m),$$

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valid under the hypotheses $O^+_{L,F} = O^+_L$, $c_0(0) = 0$ and $c_\eta(m) \in \mathbb{Z}$ for $m \leq 0$.

The quantities $\kappa_\eta(m)$ are not given in [4]; the reader is referred to [3] and [12] for their evaluation. Both go back to the computation of Kudla–Rapoport–Yang [5], whose Proposition 2.8 identifies the configurations contributing to the derivative of an incoherent Eisenstein series at the centre and asserts that *these are the only cases which contribute.*

What this note contains. Section 2 traces that assertion to its stated scope. Section 3 computes the missing local ingredient: the local Whittaker factor of the N-scaled binary lattice at a split level prime, in closed form (Theorem 3.1), with the corollary that the empirical "level support rule" of the computational literature is exactly the vanishing of that factor (Corollary 3.2). Section 4 re-runs the case analysis with this factor and finds no new case (Proposition 4.1). Section 5 records the correction that the CM values require, computed exactly, and Section 6 lists the explanations we have eliminated. Section 9 records the measured derivation of the correction's Eisenstein part: on nine bases, each preregistered against the last, it is a fixed Siegel–Weil combination of Gross genus averages whose apparent base-dependence (the "s-law") is entirely mass normalisation. Six further bases, all preregistered, then test the law where it could break: at composite s the support re-splits (and the three readings we had recorded in advance are all refuted as stated), while the doubling of the weights that accompanies the re-split turns out to be no law at all but the choice of isotropic coset at which the functional is read (Remark 9.1). Four more, at odd level, then separate the last confound and show the $N = 1$ weights following s after all. What survives across all nineteen bases, in both normalisations, is the support and a mass-ratio identity. Section 10 states what remains open.

All numerical statements below are exact rational computations, carried out in Magma; the supporting scripts are described in Appendix A.

2. The chain, and the scope of its central step

Write $U = \lambda^\perp$ for the negative definite plane attached to a CM point of discriminant d, $L_+ = L \cap \mathbb{Q}\lambda$ and $L_- = L \cap U$. Following [12, (11),(12)], one sets

$$\kappa_\eta(m) = \sum_{\mu \in L/(L_+ + L_-)} \; \sum_{x \in \eta_+ + \mu_+ + L_+} \kappa^-_{\eta_- + \mu_-}\big(m - \langle x, x\rangle/2\big),$$

where $\kappa^-_\mu(m)$ is the s-linear coefficient of the m-th Fourier coefficient $A_\mu(s, m, v)$ of an incoherent Eisenstein series of weight 1 attached to L_-, and $\kappa^-_\mu(0) = 0$ unless $\mu = 0$.

For $m > 0$ one has, with $S_{m,\mu} = \{p : p \mid \Delta \text{ or } v_p(m) > 0\}$ and $\Delta = \operatorname{disc}(L_-)$,

$$A_\mu(s, m, v) = -\frac{2\pi}{L(s+1, \chi_d)} \prod_{p \in S_{m,\mu}} \frac{W_{m,p}(s, \varphi_{\mu,p})}{1 - \chi_d(p) p^{-1-s}}, \tag{1}$$

a genuine Euler product because $\varphi_\mu = \operatorname{char}(\mu + \widehat{L_-})$ is a pure tensor. Since $A_\mu(s, m, v)$ vanishes at $s = 0$, some factor must vanish there, and differentiating (1) yields [12, Lemma 16]: the derivative is the single term at the vanishing place, and vanishes outright if two or more factors vanish.

That step is correct as an identity. What it inherits from [5] is the identification of the vanishing of a local factor with a *representability* obstruction, i.e. with membership of $\operatorname{Diff}(m)$. Kudla–Rapoport–Yang state the scope of their computation plainly:

> "in section 2, we have considered only those Eisenstein series built from sections which are defined by the characteristic functions of the completions of $\mathcal{O}_k$. More general sections could be considered at the cost of (i) more elaborate

> calculations of Whittaker functions and their derivatives used in computing the Fourier expansion of the central derivative, and (ii) incorporation of a level structure in the moduli problem." [5, §6]

(They also fix a maximal order $\mathcal{O}_B$.) The lattice L_- arising here is not of that shape. Write $\Delta = \mathrm{disc}(L_-)$. For a discriminant d with $N \nmid d$ — the case in which the level prime is *unramified* in $k = \mathbb{Q}(\sqrt{d})$, and hence, by Eichler's condition on optimal embeddings, necessarily *split* — one has $\mathrm{ord}_N(\Delta) = 2$: the binary lattice is N-scaled at the level prime, not unimodular. This is exactly a "more general section".

Remark 2.1. The distinction is invisible in the maximal-order setting: there L_- is unimodular at every split prime, no split prime can obstruct representability, and correspondingly the logarithms produced by [8, Theorem 4.1] run over ramified and inert places only. Every discriminant admitting a CM point on $X_0^D(N)$ with $N \nmid d$ has N split, so this configuration is precisely the one the maximal-order theory never meets.

3. The local Whittaker factor at a split level prime

Part (i) of the programme quoted above, at the level prime, is the following computation. Let $X = N^{-s}$.

Theorem 3.1. *Let $d < 0$ be a discriminant with $N \nmid d$ and let L_- be the negative definite binary lattice attached to a CM point of discriminant d, so $\mathrm{ord}_N(\mathrm{disc}\, L_-) = 2$. Then for every $m \geq 1$, with $j = \mathrm{ord}_N(m)$,*

$$W_{m,N}(X) \;=\; 1 + (N-1)\big(X + X^2 + \cdots + X^j\big) - X^{\,j+1}.$$

In particular

$$W_{m,N}(1) = (N-1)\,\mathrm{ord}_N(m), \qquad W'_{m,N}(1) = (j+1)\Big(\tfrac{(N-1)j}{2} - 1\Big).$$

The statement is a computation rather than a proof: it has been verified in exact arithmetic for $N = 2, 3, 5$ and all $1 \leq m \leq 60$, on the lattices L_- of $X_0^{15}(2)$, $X_0^{6}(5)$ and $X_0^{10}(3)$ at firing discriminants, giving 180 independent checks. It is pinned as a regression test in the accompanying code.

Corollary 3.2. $W_{m,N}(1) = 0$ *if and only if* $N \nmid m$.

This identifies, and explains, an empirical rule that has been observed repeatedly in computations with these curves: the correction discussed in Section 5 is supported exactly on the indices m with $N \mid m$. That rule is the vanishing of the level-prime factor.

Remark 3.3. At $j = 0$ the factor is $W_{m,N}(X) = 1 - X = 1 - N^{-s}$, the inverse local zeta factor. Its vanishing at $s = 0$ is therefore of a different nature from a representability obstruction: it holds identically for all m with $N \nmid m$, and at a split N the local quadratic space represents every m.

Remark 3.4. The value $W_{m,N}(1) = (N-1)\,\mathrm{ord}_N(m)$ coincides with the local factor Kudla–Yang obtain at $p \mid N$ for the *quaternary* Eichler lattice [6, (9.1)]. The agreement appears to be a genuine coincidence of shape between the quaternary lattice and the binary L_-; we do not have an explanation for it.

4. Re-running the case analysis

With Theorem 3.1 in hand one can redo the enumeration of [5, Prop. 2.8] for the N-scaled section. Normalising as in (1), put $f_p(s) = W_{m,p}(s)/(1-\chi_d(p)p^{-1-s})$.

Proposition 4.1. *At a firing discriminant, f_N has a simple zero at $s=0$ exactly when $N \nmid m$, and this zero is not a* Diff *zero. Consequently*

$$\operatorname{ord}_{s=0} A_\mu(s,m,v) \;=\; |\mathrm{Diff}(m)| + [\, N \nmid m\,],$$

and since incoherence forces $|\mathrm{Diff}(m)| \geq 1$:

(1) *if $N \mid m$, the order is $|\mathrm{Diff}(m)|$, and the derivative is nonzero precisely when $|\mathrm{Diff}(m)| = 1$, contributing a logarithm at the ramified or inert place of* $\mathrm{Diff}(m)$*;*
(2) *if $N \nmid m$, the order is at least 2 and the derivative vanishes.*

In neither case does a $\log N$ occur.

So the level structure only ever raises the order of vanishing: it removes contributions, and creates no new contributing configuration. The three cases of [5, Prop. 2.8] stand, with no fourth. This is confirmed computationally: over roughly 190 terms (μ, x) arising in the evaluation of $\kappa_0(m)$ on three bases, the level prime is *never* the unique vanishing place, while it lies among two or more in 60–80% of terms.

Corollary 4.2. *Within this framework, a $\log N$ can enter a CM value only through the $m = 0$ term $c_0(0)\kappa_0(0)$, where $\kappa_0(0)$ carries the summand $\sum_{p\mid N/(N,d)} \log p$ of* [12, Remark 15(2), Lemma 20].

5. The correction the CM values require

Against this stands a computation of the input form itself. Following [4, eq. (6)] one may form $F_f = \sum_{\gamma\in\Gamma_0(M)\backslash SL_2(\mathbb{Z})} (f|_{1/2}\gamma)\,\rho(\gamma^{-1})e_0$ and evaluate it numerically; the metaplectic phase is handled by evaluating both the slash and ρ along one and the same ST-word, so that they share a single metaplectic lift. Every such evaluation is gated on reproducing the principal part of F_f predicted by [4, Lemma 24] from the two scalar q-expansions of f; on the runs reported here that gate is met to 10^{-103}.

Observation 5.1. For every Borcherds form produced by the pipeline on $X_0^{15}(2)$, $X_0^{6}(5)$, $X_0^{10}(3)$ and $X_0^{21}(2)$ one finds $c_0(0) = 0$ (to 10^{-103}), so the hypothesis $c_0(0) = 0$ of [4, Thm. B] is satisfied and, by Corollary 4.2, no $\log N$ can arise.

Observation 5.2. The isotropic cosets of $L^\vee/L$ number exactly $2N-1$; the D-part of the discriminant form is anisotropic, so all of them come from the level-N hyperbolic plane. All $2N-2$ nonzero isotropic cosets carry the same constant term $c_\eta(0)$.

Observation 5.3. Let $\mathrm{mult}(f) = \frac{1}{2}c_\eta(0)$ for any nonzero isotropic η. Then adding $\mathrm{mult}(f)\cdot\sum_{p\mid N/(N,d_{\mathrm{fund}})}\log p$ to the computed CM value reproduces, exactly, the corrections independently measured by kernel-consistency on $X_0^{15}(2)$ (9 values), $X_0^{6}(5)$ (5) and $X_0^{10}(3)$ (5), and by the consistency relation of Observation 5.4 below on $X_0^{14}(5)$ (1, at eleven digits): 20 of 20. A further base $X_0^{21}(2)$ supplies nine values with no prior ground truth against which to check them.

That the correction is genuinely required, and not an artefact of a normalisation, is visible on $X_0^{6}(19)$: on a set of CM points including discriminants divisible by N, the uncorrected values are mutually *inconsistent* — the associated system has no solution — whereas with the correction the published model is recovered.

Ground truth without an Eisenstein computation. The pipeline imposes, at every rational CM point d, the consistency relation

$$\varepsilon_1(d)\,\frac{s(d)}{s(d_A)} + \varepsilon_2(d)\,\frac{\tilde{s}(d)}{\tilde{s}(d_B)} = 1, \qquad \varepsilon_i(d) \in \{\pm 1\}, \tag{2}$$

between the two Hauptmodul rows, normalised at the discriminants d_A, d_B where they vanish. Restoring a correction $r_f \log N$ to $\log|\psi_f|$ multiplies the normalised value at d by $N^{\,r_f(k(d)-k(d_{\mathrm{ref}}))}$, where $k(d) \in \{0,1\}$ records whether the term fires at d. Each rational CM point whose firing status differs from the reference's therefore contributes one equation in (N^{r_1}, N^{r_2}); two determine the pair and any further points are consistency checks. This measures the multipliers of the two Hauptmodul forms on any base the pipeline reaches, in seconds, with no Eisenstein series evaluated.

Observation 5.4. On nine bases the relation (2) determines the pair (r_{-1}, r_{-2}) uniquely:

$X_0^{15}(2)\colon (4,2)$ $X_0^{14}(5)\colon (1,1)$ $X_0^{6}(7)\colon (1,2)$ $X_0^{6}(5)\colon (0,0)$
$X_0^{10}(3)\colon (0,0)$ $X_0^{10}(11)\colon (0,0)$ $X_0^{6}(19)\colon (0,0)$ $X_0^{22}(7)\colon (0,0)$
$X_0^{34}(5)\colon (0,0)$

Every value with an independent determination is reproduced — $X_0^{15}(2)$ against the kernel-consistency measurement, $X_0^{6}(5)$ and $X_0^{10}(3)$ against the numerical evaluation of F_f, and $X_0^{6}(7)$, the one *nonzero* cross-check, against an earlier independent measurement. Conversely, on $X_0^{14}(5)$ the pair $(1,1)$ was measured first by (2) and *then* confirmed by the numerical evaluation of F_f: $\frac{1}{2}c_\eta(0) = 1.00000000000$ on the form with pole order 16, with the principal-part gate met to $2.3\cdot 10^{-37}$ and all eight nonzero isotropic cosets agreeing. This is the first verification of the rule of Observation 5.3 at $N > 2$ against ground truth obtained by an entirely independent method.

Writing $\mathrm{mult}(f) = \sum_m c(-m)A_m$ and using Borcherds' obstruction to set $A_m = -b(m)/4$ with b the relevant weight 3/2 Eisenstein coefficient, the linear functional is determined per base by the measured multipliers. Without further hypotheses the system leaves four free directions on each base; imposing the support rule of Corollary 3.2 (the weight vanishes unless $N \mid m$) removes them all, and the resulting weights are *unique* with six spare conditions satisfied on each of the three bases — eighteen independent chances to falsify the support rule, none taken. Table 1 records the result.

base	A_m (equivalently $b(m) = -4A_m$)		
$X_0^{15}(2)$	$A_2 = -1$	$A_{10} = 1$	$A_{30} = 0$
$X_0^{6}(5)$	$A_{10} = 3/2$	$A_{15} = 1/2$	$A_{30} = 3/2$
$X_0^{10}(3)$	$A_3 = 1/2$	$A_{12} = 1/2$	$A_{30} = 3/2$
$X_0^{21}(2)$	$A_2 = -2,\ A_6 = -4$	$A_{18} = 2,\ A_{42} = -8$	(0-block: $B_{21} = -5$)

Table 1. The correction, per index, as a linear functional of the principal part. The values are d-independent: at every firing discriminant the coefficient of $\log N$ in $\kappa_0(m)$ must equal A_m, whereas the recipe of §2 returns 0.

6. Explanations eliminated

Each of the following was tested against the data of Section 5 and is inconsistent with it. We record them because several are natural first guesses.

(1) mult $= c_0(0)$, or any linear combination of the two scalar constant terms $\mathrm{Coeff}(f_\infty, 0)$ and $\mathrm{Coeff}(f_0, 0)$: inconsistent on three bases.
(2) Evaluating the local Whittaker at a nonzero isotropic coset η^* rather than at 0: destroys the pattern of vanishing indices on all three bases. The mechanism is now clear: the nonzero isotropic cosets live in the level-N hyperbolic plane, so changing the coset changes *only* the level-prime factor (measured: from $1 - X^2/N$ to the constant 1, with the factors at $p \mid D$ byte-identical) and can never repair an index whose defect is at a ramified prime.
(3) Replacing the local factor at $p \mid D$ by the anisotropic branch of [6, Prop. 5.3]: does not reproduce b, under either sign of κm. (The implementation of [6, Props. 5.3, 5.4] agrees with the pipeline's own factor at every good prime, 470 checks, and at the level prime, 150 checks.)
(4) The incoherent support condition $|\mathrm{Diff}(m)| = 1$: scores 8/9 on the exact coefficients and is the only rule that predicts $b(30) = 0$ on $X_0^{15}(2)$, but is far too permissive — for $r \leq 60$ it holds at 15, 19 and 20 indices with $N \nmid r$, where the coefficient must vanish.
(5) Cancelling the zeta factor $1 - X$ at the level prime before counting vanishing places: the recovered terms differentiate at ramified and inert primes, never at N.
(6) The inner $m = 0$ of κ_η, i.e. terms with $\langle x, x\rangle/2 = m$: never occurs, in all 60 (discriminant, index) pairs over every firing discriminant of the three bases.
(7) Schofer's outer $m = 0$ term: vanishes by Observation 5.1.
(8) The re-enumeration of §4: adds no case.
(9) Any re-weighting of the existing local data: every combination of a subset of the prefactors $\{\sqrt{m}/\sqrt{|d|}$, h/w, $\prod(1 - \chi(p)/p)$, $\prod(1 - 1/p^2)^{-1}\}$ with a local rule drawn from {the product of the $W_p(1)$; the derivative of that product; the derivative taken at N, at the primes $p \mid D$, at each single prime, or at the vanishing places}, times an arbitrary constant per base, fails to reproduce the nine measured multipliers of $X_0^{15}(2)$.
(10) More strongly, *no multiplicative correction of any kind* can succeed: at two of the nine determined indices (both with $m = 10$) the product of the code's local factors vanishes while the true weight does not, so the discrepancy is infinite there. The missing contribution is a *term*, not a factor.
(11) Differentiating at the place where the ternary *space* fails to represent m (Kudla's Diff, computed at the level of quadratic spaces rather than lattice densities): reproduces $A_2 = -1$ and $A_{30} = 0$ on $X_0^{15}(2)$ exactly — the only rule found that *explains* $A_{30} = 0$ — but fails on 7 of 9 indices, and is refuted as a support rule by $X_0^6(5)$ at $m = 15$, where Diff is empty but $A_{15} = 1/2 \neq 0$. The support is $N \mid m$; whether Diff supplies the correct place to differentiate is a separate question.

7. The correction identified

The rule of Observation 5.3 reads off the constant term $c_{\eta^*}(0)$ of the input form at a nonzero isotropic coset η^*. By the constant-term pairing (pair F against a holomorphic Eisenstein series of the complementary weight 3/2 and take the constant term of the resulting weight-2 invariant), that quantity is determined by the principal part of F paired against the coefficients of E_{A,η^*}, the Eisenstein series attached to the coset η^* of the discriminant form $A = L^\vee/L$. This is precisely the object the maximal-order literature does not compute: [2], [6, §8–9] and the recipe of §2 all work with the series attached to the *zero* coset (equivalently, with sections $\mathrm{char}(\mu + \widehat{L})$ seeded at 0), and Kudla–Yang explicitly leave the general Eichler-order case open at the end of their §9.

The coefficients of $E_{A,\beta}$ for isotropic $\beta \neq 0$ are computed by Schwagenscheidt [9]. For our purposes the essential ingredient is not the main formula but the oldform relation [9, Prop. 1.4] applied to the trivial character: for $\beta = \eta^*$ of prime order $N_\beta = N$,

$$E_{A,\eta^*,\chi_0} \;=\; E_{B,0}\uparrow_B^A \;-\; E_{A,0}, \qquad B = \langle \eta^* \rangle^\perp / \langle \eta^* \rangle \;\cong\; L_1^\vee / L_1, \tag{3}$$

where L_1 is the even lattice generated by L and η^*. Pairing F against (3) and using Observations 5.1 and 5.2 — $c_0(0) = 0$, the constant $c_\eta(0)$ shared by all $2N-2$ nonzero isotropic cosets, and the vanishing of the constant-term pairing for every nontrivial character twist — gives

$$\mathrm{mult}(f) \;=\; -\frac{1}{4(N-1)} \sum_{n>0} \sum_{\gamma \in (\eta^*)^\perp} c_\gamma(-n)\, b^{B,0}_{\gamma+H}(n), \qquad H = \langle \eta^* \rangle, \tag{4}$$

in which every object on the right is a *seed*-0 Eisenstein coefficient of the smaller module B — the territory of [2], computable by the same machinery as the existing recipe, applied to L_1 in place of L.

Three structural facts corroborate the identification.

(1) The pairing constant in (4) is $4(N-1)$ — exactly the normalisation measured in Observation 5.3 (the sum of $c_\eta(0)$ over the $2N-2$ nonzero isotropic cosets equals $4(N-1)\cdot\mathrm{mult}$).
(2) $\det L_1 = \det L/N^2$: the level-prime part of L_1 is *unimodular*, so the coefficients $b^{B,0}$ have no vanishing level-prime factor. The zero $W_{m,N}(1) = 0$ at $N \nmid m$ (Corollary 3.2), which removes the majority of terms from the seed-0 pairing on A, never occurs on B — the structural mechanism by which (4) can produce the $\log N$ terms that §4 shows the seed-0 recipe cannot.
(3) $2|\det L_1|$ remains a perfect square (e.g. $2 \cdot 450 = 30^2$ on $X_0^{15}(2)$), so the discriminant and character bookkeeping of the weight-3/2 coefficient formula carries over unchanged in shape.

Two cautions, the first of which the data has since promoted to a finding. The identity (3) is proven in [9] for weight $k \geq 5/2$; at $k = 3/2$ both sides require analytic continuation, and the naive pairing argument assumes the continued series is holomorphic at the special point, where Zagier-type non-holomorphic contributions are possible. That assumption *fails*, and it fails in a way one can measure without any of the machinery of this section: for the true seed-0 series on A itself, the constant-term pairing forces $\sum_{n>0,\gamma} c_\gamma(-n)\, b^{A,0}_\gamma(n)$ to equal $-c_0(0)$ minus a spread term proportional to the multiplier — so on a form with $\mathrm{mult} = 0$ and $c_0(0) = 0$ it must vanish outright. Evaluating the left side with the naïve weight-3/2 coefficient formula (rank 3 collapses the Bruinier–Kuss local polynomials to plain representation densities, since they are evaluated where the factor $1 - p^{m-1}X$ vanishes) gives -5, not 0, on such a form of $X_0^{15}(2)$. The naïve recipe therefore violates its own constant-term identity, independently of everything specific to (4) — and this, rather than any defect of (3), is where the computation of (4) currently stops.

One natural reading of that violation — that the weight-3/2 series is merely harmonic, and the missing term is the pairing against its ξ-image, a weight-1/2 theta on A — is *refuted* by the following observation, which seems worth recording in its own right.

Proposition 7.1. *For the discriminant forms $A = L^\vee/L$ arising here (and for their rescalings $(2L)^\vee/2L$), the space of holomorphic vector-valued modular forms of weight $1/2$ for ρ_A (and for ρ_A^*) is zero.*

Proof sketch. By the Serre–Stark theorem in its vector-valued form [10, 11], every such form is a linear combination of unary theta functions; each arises from a cyclic nondegenerate orthogonal summand Z of a quotient $H^\perp/H$ of A by an isotropic subgroup H, tensored with a ρ_C-invariant vector on the complement C, and invariant vectors are supported on self-dual isotropic subgroups of C. The D-part of A is anisotropic of shape $(\mathbb{Z}/p)^2$ at each odd $p \mid D$ (Observation 5.2); an isotropic subgroup cannot meet it, a cyclic summand cannot absorb both $\mathbb{Z}/p$ factors, and passage to $H^\perp/H$ preserves it. Hence at least one anisotropic $\mathbb{Z}/p$ survives into every complement, no invariant vector exists, and every candidate theta vanishes. □

In particular $E_{A,0}$ of weight $3/2$ has no shadow, the completion reading is closed, and the input form F_f is determined by its principal part alone — a fact the computations of §8 confirm to seventy digits on pairs of forms with equal principal parts. What the failed identity signals is therefore not a non-holomorphic correction on A but a missing *channel* in the ansatz for the functional itself, which the exact evaluation of the next section locates.

8. The constant term evaluated exactly, and the two channels of the correction

The constant term that Observation 5.3 reads off admits a finite evaluation which replaces the Fourier extraction of §5 entirely. For a coset word w with matrix g and an eta monomial $\prod_d \eta(d\tau)^{r_d}$ of f, triangularising $\left(\begin{smallmatrix} d & 0 \\ 0 & 1 \end{smallmatrix}\right) g = g_d \begin{pmatrix} a_d & b_d \\ 0 & e_d \end{pmatrix}$ exhibits $(f|_{1/2}w)$ as a constant multiple of a q-series with *exact rational exponents* and coefficients that are roots of unity; the single constant per (monomial, word) is pinned numerically at one point of $\mathbb{H}$ and verified at a second, so no eta-multiplier or metaplectic bookkeeping enters. The coefficient of q^0 in $f|_{1/2}w$ is then a finite sum, and

$$c_{\eta^*}(0) \;=\; \sum_w \big(\rho(w^{-1})e_0\big)_{\eta^*}\, a_0(f|_{1/2}w)$$

is evaluated in minutes per base, with no horocycle sampling, no aliasing, and none of the cancellation constraints of §A. The same series supply the full principal part of F_f at every coset and exponent; on $X_0^{22}(3)$ this reproduces [4, Lemma 24] coefficient by coefficient to 10^{-70}, and on $X_0^{15}(2)$ the nine multipliers of Observation 5.3 are reproduced exactly, the two zero multipliers to 10^{-71}.

Two panels beyond the reach of the numerical oracle result. On $X_0^{22}(3)$ the nine multipliers are

$$(r_{-2}, r_{-1}, r_9, \dots, r_{15}) = (3,\ 3,\ -6,\ 3,\ -3,\ -9,\ 3,\ 6,\ -6),$$

and on $X_0^{21}(2)$ — a base on which no ground truth of any kind existed —

$$(2,\ 2,\ 4,\ 2,\ 2,\ 2,\ -4,\ -4,\ 0).$$

The second channel. The nine $X_0^{22}(3)$ values are *inconsistent* with any linear functional of the principal part supported on the indices $N \mid m$: the combinations of forms $(f_{-2} - f_9 + f_{11})$ and $(-f_{-1} - f_9 + f_{12})$ cancel every N-divisible index of the principal parts identically, yet their multipliers sum to 6 and -6. The residual support of both combinations is the set $m \in \{1, 4, 16, 25\}$ — perfect squares — with residual coefficients $(4,4,2,2)$ and $(-4,-4,-2,-2)$, so both reduce to the single equation $12\, w_\square = 6$. The correction therefore carries a second channel:

$$\mathrm{mult}(f) \;=\; \underbrace{\sum_{N \mid m} c_0(-m)\, w(m)}_{\text{level channel}} \;+\; w_\square(D,N) \sum_{k \geq 1} c_0(-k^2), \tag{5}$$

with $w(m)$ depending on m only through the square class of m/N. With both channels present, the measured multipliers of *every* base are consistent, with spare conditions throughout; the data force

$$w_\square = 0 \text{ on } X_0^{15}(2),\ X_0^6(5),\ X_0^6(13),\ X_0^6(17); \qquad w_\square = \tfrac{1}{2} \text{ on } X_0^{22}(3); \qquad w_\square = 1 \text{ on } X_0^{21}(2).$$

The square-class sum is the signature of the *scalar* Zagier weight-3/2 structure — which is consistent with Proposition 7.1: no vector-valued shadow exists on A, but the scalar theta enters through the cusp expansions of f itself. The bases with $w_\square = 0$ are exactly those on which the level-support rule appeared exact, which is why this channel escaped notice through eighteen bases of fitted corrections.

Conjecture 8.1. *$w_\square(D,N) \neq 0$ if and only if no prime of DN splits in $\mathbb{Q}(i)$ — the embedding criterion for the orders of discriminant $-4k^2$ — and in that case $w_\square = 1/(N-1)$.*

The condition matches all six determined values above (DN contains a prime $\equiv 1 \bmod 4$ precisely on the four bases with $w_\square = 0$), and the magnitude matches $N = 2, 3$. We record, in advance of the measurements, the values the conjecture predicts on the panels currently being computed: 1 on $X_0^{33}(2)$; $\frac{1}{2}$ on $X_0^{38}(3)$ and $X_0^{14}(3)$; 0 on $X_0^{35}(2)$, $X_0^{22}(5)$ and $X_0^{34}(5)$; $\frac{1}{6}$ on $X_0^{22}(7)$; $\frac{1}{10}$ on $X_0^6(11)$.

Scorecard at the time of writing. The panels for $X_0^{33}(2)$, $X_0^{35}(2)$, $X_0^{22}(5)$, $X_0^{38}(3)$, $X_0^{14}(3)$, $X_0^6(11)$, $X_0^{10}(11)$, $X_0^{14}(5)$, $X_0^6(19)$, $X_0^{34}(5)$ and $X_0^{22}(7)$ have since been measured. Every one is consistent with the two-channel form (5), with spare conditions; two of the preregistered predictions have been decided, and both *hit*: $X_0^{14}(5)$ pins $w_\square = 0$, and $X_0^{34}(5)$ pins $w_\square = 0$ at rank 7 of 9 (both have $5 \mid DN$ split in $\mathbb{Q}(i)$); the latter panel also pins the k-twist values $w(1,1) = w(1,2) = w(1,4) = 0$, $w(17,1) = 0$ and $w(2,1) = w(34,1) = \frac{1}{2}$. On the remaining bases the nine-form panel leaves $w_\square$ undetermined: the square-class column of the principal-part profile matrix lies in the span of the level-channel columns, so no value of $w_\square$ is forced either way. This aliasing is *intrinsic* rather than an accident of the panel — see the assembly analysis below — so the outstanding predictions must be decided by the derivation, not by further panels of the same kind.

The level channel. Within the first channel the data pin two further structural facts. The weight at $m = Ndk^2$ (d squarefree) depends on k through a sign: on $X_0^{10}(7)$ the two Hauptmodul rows force $w(7) = +\frac{1}{2}$ and $w(28) = -\frac{1}{2}$, and across all bases the k-dependence is consistent with the twist $\lambda(k)\left(\frac{k}{N}\right)$ (Liouville times Legendre), while k-independence and k-linearity are refuted. And no assignment of the weights by a fixed product of Legendre symbols in (D, N, d) covers all bases: three successive such laws, each consistent with all data available when it was formulated, were refuted by the next base measured ($X_0^6(13)$, then $X_0^{10}(7)$, then $X_0^{22}(3)$ and $X_0^{38}(3)$). The surviving magnitudes $0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}$ vary with the base in the manner of representation numbers rather than characters. Since the coset sum above evaluates $c_{\eta^*}(0)$ as a finite sum of Weil-representation entries — quadratic Gauss sums — against the cusp expansions of f, grouping it by cusp class is a finite, exact route to the closed form; we have executed it, with the following results.

The cusp-class assembly. Throughout this subsection $A = L^\vee/L$ carries the *negated* quadratic form $\bar{q} = -q$ (the dual Weil representation of the evaluation above is the representation $\rho_{\bar{A}}$ of [7] attached to $\bar{A} = (A, \bar{q})$; the identification is verified coset by coset in `rho5.m`), η^* denotes a nonzero isotropic element, M is the lattice level, and for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with lower row (c, d) we write $g = \gcd(c, M)$ for its cusp class. We assume DN even, the case of all computations in this paper.

Proposition 8.2 (local structure)**.** *The Jordan components of $\bar{A}$ are as follows. For each odd $p \mid D$ the p-part is the anisotropic plane $(\mathbb{Z}/p)^2$; for each odd $p \mid N$ it is the hyperbolic plane; the* 2*-part has order* 8 *and equals $A_2^1 \oplus C_2$ if $2 \mid N$, and $A_2^1 \oplus B_2$ if $2 \mid D$, in the notation of* [7, §2]*. Consequently the Gauss function of the* 2*-part on odd arguments is*

$$G_2(x) \;=\; \epsilon_D \left(\tfrac{2}{x}\right) e_8(x), \qquad \epsilon_D = \begin{cases} +1, & 2 \mid N, \\ -1, & 2 \mid D, \end{cases}$$

by [7, Lemmas 3.7, 3.8] *(the B_2 factor contributes the constant $\left(\frac{3}{2}\right) = -1$).*

Proof. The completion $L \otimes \mathbb{Z}_p$ depends, up to isometry, only on the local type of the order: at $p \mid D$ the maximal order of the ramified quaternion algebra over $\mathbb{Q}_p$ is unique up to conjugation, and at $p \mid N$ so is the Eichler order of level p in $M_2(\mathbb{Q}_p)$. Hence the p-part of the discriminant form is a single isomorphism class determined by the case, and one computation per case identifies it: the odd parts are the standard anisotropic (ramified) and hyperbolic (split Eichler) planes, and for $p = 2$ the $\bar{q}$-value multisets $\{0, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ $(2 \mid N)$ and $\{0, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, \frac{3}{4}\}$ $(2 \mid D)$, computed in `g2jordan.m` and constant across ten bases, identify $A_2^1 \oplus C_2$ and $A_2^1 \oplus B_2$ respectively. □

Theorem 8.3 (the ρ-entry at a cusp class)**.** *Let γ have lower row (c, d) with class g, and let $\widetilde{\gamma}$ be its canonical metaplectic lift. Then $(\rho_{\bar{A}}(\widetilde{\gamma}) e_0)_{\eta^*}$:*

1. *vanishes iff $N \mid g$;*
2. *has absolute value $g/\sqrt{|A|}$ for odd g and g/M for $4 \mid g$;*
3. *for odd g and the canonical representative $(a = d = 1,\ c = -g)$ equals $(g/\sqrt{|A|})\, e_8(E_0(g))$ with*

$$E_0(g) \;=\; 2 + 4\, r(g) - 4\, \omega(g) + \arg_8 G_2(-g) \mod 8, \qquad \arg_8 G_2(x) = x + 4 \left[\left(\tfrac{2}{x}\right) = -1\right],$$

   where $r(g) = \#\{p \mid g : p \equiv 3 \bmod 4\}$ and $\omega(g)$ the number of prime divisors; in particular E_0 is independent of the base (D, N);
4. *for $4 \mid g$ and the canonical representative it is real, of sign $\mu(g/4)$.*

Proof. By [7, Thm. 6.4], for $x = 0$ the η^*-component of $\rho_{\bar{A}}(\widetilde{\gamma}) e_0$ is the *single* term $\xi(a, c)\sqrt{|A[c]|/|A|}\; e(ac\, \bar{q}(y_0) + B(x_c, y_0))$ if $\eta^* \in cA + x_c$ (with $cy_0 + x_c = \eta^*$), and 0 otherwise. (i): the element x_c is 2-adic, and for $N \mid c$ the N-part of cA is trivial while η^* has nonzero N-part, so the support condition fails; conversely for $N \nmid g$ it holds. (ii): $|A[c]| = \prod_{p \mid g \text{ odd}} p^2$ times 8 if $4 \mid c$, and $|A| = M^2/2$. (iii): choose y_0 inside the N-part; then $\bar{q}(y_0) = c^{-2} \bar{q}(\eta^*) = 0$ and (for odd c) $x_c = 0$, so the phase is exactly $\xi(a, c)$, which [7, Def. 6.1] factors as $e_4(-\text{sign})\, \xi_0\, \xi_2 \prod_J \xi(J)$ over the Jordan components of Proposition 8.2. At the canonical representative: $\xi_2 = 1$; $\xi_0 = \left(\frac{-1}{-g}\right)(-1, -g)_\infty = e_8(4 + 4r(g))$; each anisotropic pair with $p \nmid g$ contributes $\left(\frac{t_1 t_2}{p}\right) e_8(2 - 2p) = -1$ and each hyperbolic pair $+1$ ([7, Lem. 3.6], independently of the argument); the components at $p \mid g$ contribute 1; and the 2-part contributes $G_2(-g)$ of Proposition 8.2. Collecting, and using $e_4(-\text{sign}(\bar{A})) = e_8(-2)$ together with the two Gauss-sum factors $e_8(-3)$ and $e_8(1)$ arising from the normalisation of [7, eq. (6.6)] (the latter because the auxiliary $d' \equiv 1 \bmod 8$ forces its Kronecker symbol to $+1$), gives $E_0(g) = 2 + 4r - 4\omega(g) + \arg_8 G_2(-g) + 4\, \omega_{\text{odd}}(D) + 4\, [2 \mid D]$. The last two terms cancel: an indefinite quaternion algebra ramifies at an even number of finite primes, so $\omega_{\text{odd}}(D)$ is odd exactly when $2 \mid D$, which is exactly when $\epsilon_D = -1$ shifts $\arg_8 G_2$ by 4. This proves (iii) and the base-independence. (iv): for $4 \mid c$ the level-4 component is annihilated, $x_c = 0$ again, and every factor of ξ is real ($\xi_2 = e_8(-2(c_2 - 1 + s_2))$ with c_2 the odd part of c and s_2 the 2-part signature contributes a sign, as do ξ_0 and the anisotropic pairs prime to g); the product of signs is $\mu(g/4)$ by the same bookkeeping. □

The theorem reproduces all nine measured odd classes ($g = 1, 3, 5, 7, 11, 15, 21, 33, 35$ across six bases) and all four measured $4 \mid g$ classes. On $\omega(g) \le 2$ — every class measurable on these bases — E_0 coincides with the symbol form $1+6r(g)+4s(g)$, $s(g) = \#\{p < q \mid g : (\frac{p}{q})(\frac{q}{p}) = +1\}$, but the two diverge from $\omega(g) = 3$ onward: the symbol form is only the low-complexity shadow of E_0 (`rho5.m`, `thetag-derivation.md`). Word-based evaluations differ from the canonical lift by the metaplectic centre (± 1 per word, the Kubota cocycle product of the letters), which cancels in the pairing below.

Proposition 8.4 (class constancy). *Let f be weakly holomorphic of weight $1/2$ for $\Gamma_0(M)$ whose multiplier restricted to $\Gamma_0(\mathrm{lev}(\bar{A}))$ equals the character χ by which that group acts on e_0 (the case of all Borcherds inputs here, forced by well-definedness of the coset average of* [4, Lem. 24]*). Then $(\rho(w^{-1})e_0)_\eta\, a_0(f|_{1/2}w)$ is, for every isotropic η, constant as w ranges over a cusp class, and consequently*

$$c_{\eta^*}(0) \;=\; \sum_g N_g \left(\rho(w_g^{-1})e_0\right)_{\eta^*} a_0(f|_{1/2}w_g)$$

with one arbitrary coset w_g per class and N_g the number of cosets in the class.

Proof. Replacing w by $\gamma w T^k$ with $\gamma \in \Gamma_0(M)$ multiplies $a_0(f|_{1/2}w)$ by $\chi(\gamma)$ (the T-shift fixes the q^0 coefficient), while e_0 is a $\rho(\Gamma_0(\mathrm{lev}))$-eigenvector with character χ ([7, Lem. 5.5, 5.6]; verified to 10^{-59} in `rho6.m`), so the ρ-side acquires $\overline{\chi(\gamma)}$, and the isotropy of η makes its component T-invariant. The factors cancel. $\square$

The constancy is verified on 570 class/form/base checks (`cuspclass2.py`), and Theorem 8.3 with Proposition 8.4 make the evaluation of $c_{\eta^*}(0)$ a sum over *classes* rather than cosets — the form in which it is deployed in production (`MOMultiplierExact`), with the constancy re-verified at runtime on sampled cosets. The remaining ingredient, $a_0(f|_{1/2}w_g)$, is itself closed-form: the pinned constants of the evaluation equal $\zeta_w \prod_d \big[\varepsilon(g_d)\, e(b_d/24e_d)\, e_d^{-1/2}\big]^{r_d}$ with ε the Dedekind-sum eta multiplier and ζ_w the metaplectic cocycle unit of the word, certified on all 144 cosets of $X_0^{15}(2)$ and all 288 of $X_0^{10}(7)$ (`cusp4.m`), so nothing in the assembly remains numerical.

Because $c_{\eta^*}(0)$ is linear in f, the assembly can be probed one eta monomial at a time — each monomial of the $X_0^{15}(2)$ forms is itself a weakly holomorphic weight-$1/2$ form of the same multiplier character, holomorphic at every intermediate cusp, and there are 86 of them, extended to 158 by monomials appearing in no form of the panel (`cusp5.m`–`cusp7.m`). On this space the functional does *not* factor through the ∞-side principal part (residual 2.08): the 0-side principal part genuinely enters, and the joint $(\infty, 0)$-side factorisation holds to 10^{-11} with eighty independent quotient conditions. The ledger's "0-block weight 0" finding, and the ∞-only shape of (5) itself, are therefore *representatives* of the true functional modulo the relation ideal of the form space: the joint principal-part coordinates of this character's forms span only 78 of the 128 channel slots, and the 50 leftover combinations are identities of the space, unremovable by further probes. This is the structural origin of the $w_\square$ aliasing reported above, and it sharpens what the derivation must produce: the functional as an element of the quotient, with the square-class channel as its panel-span shadow.

9. The Eisenstein part as ternary theta arithmetic: the s-law and its Siegel–Weil origin

The assembly of the previous subsection makes the functional a finite exact pairing against the cusp constants of f; the derivation it calls for has since been carried out in measured stages, each preregistered before the next base was computed. We record here the state of that campaign. Throughout, $D = qs$ with $q > s$ its two prime factors, and $\mathrm{Gr}(D', R)$ denotes the *genus* of the ternary lattice $\{\alpha \in \mathcal{O}' : \operatorname{tr}\alpha = 0\}$ with the reduced-norm form, $\mathcal{O}'$ an Eichler order of level R in the *definite* quaternion algebra of discriminant D' (for $D'R = DN$ this is the Jacquet–Langlands partner of (D, N)); $\bar\theta(D', R)$ is its mass-weighted average theta series and $\Theta(D', R) = \text{mass} \cdot \bar\theta$ the corresponding genus sum.

Remark 9.1 (the tracked coset: what is arithmetic and what is normalisation)**.** Every fit below reads the ρ-vector of Theorem 8.3 at *one* isotropic coset, and that choice is a normalisation which must be named. At $N > 1$ the coset is a nonzero η^* — the functional of Observation 5.3, well defined because all $2N - 2$ choices agree (Observation 5.2); at $N = 1$ there is no such coset, since $2N - 1 = 1$ leaves only $\eta = 0$, so the $N = 1$ bases fit a different linear functional. This is the *only* difference between the two regimes. Forcing the zero coset at an $N > 1$ base reproduces the entire $N = 1$ shape — the third term drops, the remaining pair doubles, and the weight sum moves from 0 to 1:

base	fit of E_{eis} mod cusp	$\sum w$
$X_0^{15}(2),\ \eta^*$ (default)	$-\frac{1}{2}\bar\theta(5,2) + \bar\theta(5,6) - \frac{1}{2}\bar\theta(30,1)$	0
$X_0^{15}(2),\ \eta = 0$	$-\bar\theta(5,2) + 2\,\bar\theta(5,6)$	1
$X_0^{22}(3),\ \eta^*$ (default)	$-\bar\theta(11,3) + \frac{3}{2}\bar\theta(11,6) - \frac{1}{2}\bar\theta(66,1)$	0
$X_0^{22}(3),\ \eta = 0$	$-2\,\bar\theta(11,3) + 3\,\bar\theta(11,6)$	1

At $X_0^{22}(3)$ the zero-coset fit is literally $(-2, +3)$, the weights of every $N = 1$ base recorded below. In particular the third term does not vanish at $N = 1$ for the reason one would guess: at $X_0^{15}(2)$ the slot $(30, 1)$ is definite, of one class, and it still drops under $\eta = 0$. Two consequences, which we apply throughout. Absolute weights are comparable only *within* a regime; the weight formulas of Observation 9.3 are those of the η^* normalisation. And the convention-free content of these fits is the mass *ratio* of Observation 9.5, unchanged by any rescaling — which is why it, alone among the readings, survives across both regimes.

Observation 9.2 (the weight-3/2 residue pairing)**.** On every base measured, the class-assembled ρ-vector of Theorem 8.3 lies in the span of the 0-cusp constant vectors of the holomorphic weight-3/2 eta monomials with the conjugate panel character (residuals $\sim 10^{-42}$ at working precision 80), so the functional is represented by a weight-3/2 form E^*; its Eisenstein component $E_{\text{eis}} \in M_{3/2}(\Gamma_0(4DN))$ is exactly rational. Three constants are shared by all nine bases of Observation 9.3: the Eisenstein dimension of $M_{3/2}(\Gamma_0(4DN))$ is 11, the rank of the constants map is 7, and the aliasing kernel of the Gross-average fit has dimension 6. They are constants of that family of levels rather than of the phenomenon: the aliasing is 25 at $X_0^{210}(1)$ and $X_0^{330}(1)$; the triple degenerates to $(5, 3, 1)$ at the four small bases $D = 2q$, $N = 1$ of Observation 9.7 (levels 40–104); and the odd-level family of Observation 9.8 has its own uniform pair, Eisenstein dimension 7 and constants-rank 7, at all four of its levels $60, 84, 132, 140$.

Observation 9.3 (the s-law; nine bases, all with $N > 1$)**.** On every base, in the η^* normalisation of Remark 9.1, E_{eis} lies in the span of the $\bar\theta(D', R)$ over the definite structures with $D'R \mid DN$, modulo cusp forms, with support $\{(q, N), (q, sN), (DN, 1)\}$ and weights

$$w_1 = -\frac{1}{s-1}, \qquad w_2 = \frac{\psi(s)}{2\varphi(s)} = \frac{s+1}{2(s-1)}, \qquad w_3 = -\frac{1}{2},$$

depending on the base only through s. The nine bases are $X_0^{15}(2)$, $X_0^{21}(2)$, $X_0^{33}(2)$ ($s = 3$: weights $-\frac{1}{2}, 1, -\frac{1}{2}$); $X_0^{22}(3)$, $X_0^{34}(3)$, $X_0^{10}(7)$ ($s = 2$: $-1, \frac{3}{2}, -\frac{1}{2}$); $X_0^{35}(2)$, $X_0^{55}(2)$ ($s = 5$: $-\frac{1}{4}, \frac{3}{4}, -\frac{1}{2}$); and $X_0^{77}(2)$ ($s = 7$: $-\frac{1}{6}, \frac{2}{3}, -\frac{1}{2}$). The law was found on the first six and *preregistered* on the last three, each of which extended its range ($s = 3$ at a new q; $s = 5$ at a new q; $s = 7$ new): all three hit. Two simpler laws were killed by preregistered bases along the way: weights following N alone (killed by $X_0^{35}(2)$), and weights following $r = DN/q$ (killed by $X_0^{10}(7)$, which also fixed q as the largest *ramified* prime: there $N = 7 > q = 5$ and the support remains at $(5, *)$).

Observation 9.4 (Siegel–Weil identification)**.** On $X_0^{15}(2)$ the identification is complete. The 16 ternary lattices whose thetas span the relevant subspace of $M_{3/2}(\Gamma_0(60))$ fall into 13 genera, and each genus in the support is quaternionic: the assignment $(D', R) \mapsto \mathrm{Gr}(D', R)$ sends the four definite factorizations $30 = D'R$ bijectively onto the four genera of discriminant 900, distinguishes same-discriminant genera by which prime ramifies (e.g. $\mathrm{Gr}(2, 5)$ and $\mathrm{Gr}(5, 2)$ are the two genera of discriminant 100), and is inert under level structure at an already-ramified prime. The independent decomposition of E_{eis} over the 13 genus averages plus cusp forms reproduces the fitted weights term by term: $E_{\mathrm{eis}} = -\frac{1}{2}\,\bar\theta(5, 2) + \bar\theta(5, 6) - \frac{1}{2}\,\bar\theta(30, 1)$ modulo cusp forms. The s-law weights are coefficients of genuine Siegel–Weil genus averages, not of single-class thetas.

Observation 9.5 (the mass identities; the s-dependence is normalisation)**.** On all nine bases the genus masses of the support obey

$$\mathrm{mass}(q, N) = \frac{(q-1)(N+1)}{96}, \qquad \frac{\mathrm{mass}(q, sN)}{\mathrm{mass}(q, N)} = \frac{s+1}{2}, \qquad \mathrm{mass}(DN, 1) = \frac{(q-1)(s-1)(N-1)}{192},$$

so that, over the genus *sums*, the s-dependence of the weights disappears:

$$E_{\mathrm{eis}} \;=\; \frac{96}{(q-1)(s-1)(N+1)}\Big[-\Theta(q, N) + \Theta(q, sN) - \frac{N+1}{N-1}\,\Theta(DN, 1)\Big] \quad \text{mod cusp forms,}$$

an identity verified on all nine bases (in the η^* normalisation); the second factor of the middle mass ratio is multiplicative, $\prod_{p|s}(p + 1)/2$, over the primes of s. The bracket depends on the base only through N, and the fitted "s-law" is exactly the mass normalisation of a fixed Siegel–Weil combination.

The scale-free part of this,

$$\left|\frac{w_2}{w_1}\right| \;=\; \frac{\mathrm{mass}(q, sN)}{\mathrm{mass}(q, N)},$$

is the one statement here untouched by Remark 9.1, and it is the widest-held: it holds on all *nineteen* bases measured — the nine above, the two composite-D bases, the four $N = 1$ bases and the four odd-level bases below — in both normalisations, in both parities of DN, and, where the support is degenerate, on both competing supports simultaneously (Observations 9.7 and 9.8). It is therefore the form in which we expect these fits to enter a Siegel–Weil derivation.

Remark 9.6 (preregistration: the first composite s)**.** The base $X_0^{210}(1)$ ($q = 7$, $s = 30$, $N = 1$) separates three extensions of the law that agree on all nine prime-s bases, and we record their predictions in advance of the measurement, which is in flight at the time of writing. (A) The reciprocal reading of Observation 9.3 verbatim: $w_1 = -\frac{1}{29}$, $w_2 = \frac{31}{58}$, $w_3 = -\frac{1}{2}$. (B) The ψ/φ reading (ψ the Dedekind psi): $w_1 = \frac{1}{2} - \psi(30)/2\varphi(30) = -4$, $w_2 = \psi(30)/16 = \frac{9}{2}$. (C) The mass reading of Observation 9.5: $\mathrm{mass}(7, 30)/\mathrm{mass}(7, 1) = 9 = \prod_{p|30} \frac{p+1}{2}$, hence $w_1 = -\frac{1}{29}$, $w_2 = +\frac{9}{29}$, and *no third term*: the structure $(210, 1)$ is indefinite (four ramified places would be required), and its formal mass $(q-1)(s-1)(N-1)/192$ vanishes at $N = 1$. Reading (C) is the only one coherent about the third term; readings (A) and (C) agree on w_1. The discriminating ratio is $|w_2/w_1| = \frac{31}{2},\ \frac{9}{8},\ 9$ respectively.

Scorecard: measured, same day. The verdict refutes all three readings *as stated*, in an instructive way: their shared support assumption was wrong. The measurement (residual $1.0 \cdot 10^{-42}$, rank 100 of 100, exact rational recognition) gives

$$E_{\text{eis}} \;=\; -2\,\bar\theta(105,1) + 3\,\bar\theta(105,2) \quad \text{mod cusp forms,}$$

and the support is genuine, not a pivot artifact of the 25-dimensional aliasing: E_{eis} lies in the span of $\{\bar\theta(105,1), \bar\theta(105,2)\}$ plus cusp forms, and does *not* lie in the span of $\{\bar\theta(7,1), \bar\theta(7,30)\}$ plus cusp forms (rank 87 against 88). Thus at a four-prime discriminant the law re-splits $D = 210$ as $q = 105$ — the full odd ramified part, itself a definite discriminant — and $s = 2$: the weights are twice the prime-s law at $s = 2$, the mass-ratio law of Observation 9.5 extends to ten bases ($\text{mass}(105,1) = \frac{1}{4}$, $\text{mass}(105,2) = \frac{3}{8}$, ratio $\frac{3}{2} = |w_2/w_1|$), and the third term is absent exactly as reading (C) predicted — though for the wrong reason: (C) attributed the absence to the indefiniteness of $(210,1)$, and Remark 9.1 exhibits a *definite* third slot, $(30,1)$ at $X_0^{15}(2)$, dropping under the same normalisation. The overall scale (-2 against the prime-case $-1/(s-1) = -1$) looked at this point like a second law to be found; Remark 9.1 shows it is not arithmetic at all — $X_0^{210}(1)$ has $N = 1$, hence is normalised at $\eta = 0$, and the same doubling appears at $N > 1$ the moment that coset is forced. What does remain a constraint on the derivation is the composite-D' genus mass, which carries a factor $2^{\omega(D')-1}$ relative to the multiplicative prime-case formula of Observation 9.5.

Observation 9.7 (the eleventh base, and the four $N = 1$ bases)**.** Five further measurements, in two preregistered groups, complete the picture. (i) $X_0^{330}(1)$ ($q = 11$, $s = 30$; $M = 660$, residual $1.7 \cdot 10^{-42}$) hits all four predictions recorded in advance — support $\{(165,1), (165,2)\}$ with $q = D/2$ the odd ramified part and no third term, weights $(-2, +3)$, mass ratio $\frac{3}{2}$ from the advance-computed $\text{mass}(165,1) = \frac{5}{12}$, $\text{mass}(165,2) = \frac{5}{8}$ (the same $2^{\omega-1}$ factor), and rank 15 with aliasing 25. Its falsifier fired: the competing largest-prime support $\{(11,1), (11,30)\}$ is excluded (rank 135 against 136), while the winner carries E_{eis} with aliasing dimension 0 — the weights are pinned with no gauge freedom at all. We record that 330 cannot separate "odd part" from "largest definite divisor": those coincide on every even D, so $X_0^{330}(1)$ replicates the $X_0^{210}(1)$ re-split at fresh primes rather than testing it. (ii) Composite q and $N = 1$ are perfectly confounded across all eleven bases above; the missing cell is prime q with $N = 1$, i.e. $D = 2q$ at $M = 20, 28, 44, 52$. At $q = 5, 7, 11, 13$ (residuals $\sim 10^{-44}$) one finds $E_{\text{eis}} = -2\,\bar\theta(q,1) + 3\,\bar\theta(q,2)$ modulo cusp forms, identical to the composite bases: the doubling does not track composite q. Here the support is genuinely degenerate — the competitor $\{(2,1), (2,q)\}$ also carries E_{eis}, with weights $\left(-\frac{2}{q-1}, \frac{q+1}{q-1}\right)$ — and the mass-ratio law holds on *both* supports at all four bases ($\frac{3}{2}$ on the canonical one; $3, 4, 6, 7$ on the competitor, matching its masses), so it does not depend on the choice of q as the largest ramified prime.

These six $N = 1$ bases are what forced Remark 9.1: the doubling they exhibit survived two successive explanations (composite D, then $N = 1$ itself) before the direct test — forcing $\eta = 0$ at $X_0^{15}(2)$ and $X_0^{22}(3)$ — identified it as the tracked coset and dissolved it. The surviving arithmetic is the support (including the re-split $q = D/2$ at four-prime D) and the mass-ratio law.

One confound remained. All six of those bases have $s = 2$ — the four small ones outright, $X_0^{210}(1)$ and $X_0^{330}(1)$ after the re-split — so "the $N = 1$ weights are the constant $(-2, +3)$" and "the $N = 1$ weights are twice the s-law" agree on every one of them and nowhere else. Separating them needs $N = 1$ with $s > 2$, hence odd D, hence odd DN: a regime with level $4DN$ rather than $2DN$, which the computation had never entered. It turns out to require only that the 2-part of the discriminant form, there a single $\mathbb{Z}/2$ of odd type rather than $(\mathbb{Z}/2)^3$, be admitted into the closed form of Theorem 8.3; the four smallest such bases are then cheap.

Observation 9.8 (the odd-level quartet; the $N = 1$ weights follow s)**.** At $X_0^{15}(1)$, $X_0^{21}(1)$, $X_0^{33}(1)$ ($s = 3$) and $X_0^{35}(1)$ ($s = 5$), all preregistered, the residue pairing holds (residuals $8.0 \cdot 10^{-42}$, $6.1 \cdot 10^{-41}$, $3.2 \cdot 10^{-42}$, $9.7 \cdot 10^{-43}$) and

base	E_{eis} mod cusp	$\lvert w_2/w_1\rvert$
$X_0^{15}(1)$	$-\bar\theta(5,1) + 2\,\bar\theta(5,3)$	2
$X_0^{21}(1)$	$-\bar\theta(7,1) + 2\,\bar\theta(7,3)$	2
$X_0^{33}(1)$	$-\bar\theta(11,1) + 2\,\bar\theta(11,3)$	2
$X_0^{35}(1)$	$-\frac{1}{2}\bar\theta(7,1) + \frac{3}{2}\bar\theta(7,5)$	3

each restricted support carrying E_{eis} with aliasing dimension 0. These are $2\big(-\frac{1}{s-1}, \frac{s+1}{2(s-1)}\big)$ exactly, so the $N = 1$ weights follow s and the constant reading is refuted. Two further facts, neither predicted. First, the support is *degenerate*: the other ramified prime also carries E_{eis}, at aliasing dimension 0, with weights obeying the same formula read with its own (q', s') — $(3,5) \mapsto (-\frac{1}{2}, \frac{3}{2})$, $(3,7) \mapsto (-\frac{1}{3}, \frac{4}{3})$, $(3,11) \mapsto (-\frac{1}{5}, \frac{6}{5})$, $(5,7) \mapsto (-\frac{1}{3}, \frac{4}{3})$. The law is thus not attached to a choice of q at all when D has two primes. Second, the mass-ratio identity holds on all eight of these fits, canonical and competing, from masses computed in advance; it now stands at *nineteen* bases, across both normalisations, both parities of DN, and every support that carries E_{eis}.

The degeneracy is worth isolating, because it locates where the re-split rule has content. A two-prime discriminant cannot select a support: either factorisation works, and the choice "q = the larger ramified prime" is a convention. A four-prime discriminant does select — at $X_0^{210}(1)$ and $X_0^{330}(1)$ the competing span genuinely fails the rank test. This is why the rule is tested only at four primes, and why the case that would distinguish "odd part" from "largest definite divisor" must have all four primes odd: the minimal such base is $D = 1155$, where the odd part is the whole of D and hence indefinite, so the two candidate rules diverge for the first time.

10. The open question

Corollary 4.2 and Observation 5.1 are in direct tension with Observation 5.3 and with the failure of the uncorrected computation on $X_0^6(19)$. We see two possibilities and cannot presently distinguish them.

1. The constant term $c_0(0)$ of the form F actually attached to ψ_F differs from that of the coset sum F_f. The principal parts agree to 10^{-103}, but the principal part alone does not determine $c_0(0)$.
2. The correction compensates an error elsewhere in the computation of the CM values, outside Schofer's formula. It is a d-independent linear functional of the principal part, which is the shape of a per-form normalisation.

The identification of §7 sharpens the second reading. The $m = 0$ term of the formula, as transmitted through [8, 12], collapses the sum $\sum_\eta c_\eta(0)\kappa_\eta(0)$ to the single term $c_0(0)\kappa_0(0)$ via a condition of the form $\delta_{0,\mu}$. But the collapse condition constrains only the *negative-definite* part of the coset, so a nonzero isotropic η with $\eta_- = 0$ is not excluded by it — and by Observations 5.2 and 5.3 it is exactly those cosets, with their shared nonzero constant term, that carry the correction. On this reading nothing outside Schofer's formula is being compensated; rather, terms inside it were dropped.

In either case Theorem 3.1 and Proposition 4.1 stand, and part (ii) of the Kudla–Rapoport–Yang programme — the incorporation of a level structure — remains to be carried out. Table 1 is what any corrected formula must reproduce, index by index.

A third reading should be recorded, because it is not excluded by anything above. The evaluation of $\kappa_\mu^-(m)$ for $m > 0$ has never been checked against an independent computation at $N > 1$ — neither here nor, as far as we can determine, in the literature. Yang's worked examples [12, Ex. 21–24] take $B = \left(\frac{-1,3}{\mathbb{Q}}\right)$ with a *maximal* order, and indeed $L = L_+ + L_-$ there; the corresponding regression tests in the accompanying code likewise cover only $N = 1$. What Observation 5.4 now supplies is a verification at $N > 1$ of the *correction rule* — the multiplier $\frac{1}{2}c_\eta(0)$, confirmed at $N = 5$ against ground truth measured by an independent method — but the evaluation of $\kappa_\mu^-(m)$ itself, which sits upstream in the $m > 0$ terms, remains unverified at $N > 1$.

Remark 10.1 (Published CM values need not discriminate)**.** A natural-seeming check is to compare against a published CM value on a base with $N > 1$ and a nonvanishing correction; $X_0^{14}(5)$ with $s(\tau_{-280}) = 5/16$ [4, Ex. 36] appears to be the only available instance. It cannot work. The Hauptmodul there is normalised by $s(\tau_{-4}) = \infty$, $s(\tau_{-11}) = 1$, $s(\tau_{-35}) = 0$, so the published value depends only on the ratio $s(-280)/s(-35)$ — and -35 and -280 are both non-firing ($5 \mid d$), so any correction multiplies both by the same power of N and cancels; the remaining anchors sit at 0 and ∞, where a multiplicative factor is invisible. Indeed the pipeline reproduces 5/16 exactly both with and without the correction. A published value can only test the correction if its normalising anchors and the tested point lie on opposite sides of the firing rule; this should be checked, at a cost of nothing, before any such comparison is attempted. The consistency relation (2), by contrast, is sensitive exactly where [4, Ex. 36] is blind, and is what supplied the $X_0^{14}(5)$ ground truth here.

Appendix A. Reproducibility

The computations were carried out in Magma against the codebase accompanying this note.

- `VectorValuedForm.m` constructs F_f: coset representatives for $\Gamma_0(M)\backslash SL_2(\mathbb{Z})$, ST-words, the weight 1/2 metaplectic slash, the Weil representation over $\mathbb{C}$, and the Fourier extraction. The action of $\rho(S)$ is implemented as a finite Fourier transform on $L^\vee/L$, which reduces the cost from $|G|^2$ to $|G|\sum_r d_r$ and the memory from $|G|^2$ to $|G|$; without this the case $|G| = 24200$ is out of reach.
- `EisensteinLocalFactors.m` implements [6, (2.16),(2.17), Props. 5.3, 5.4] and exposes `LocalWhittakerPolynomial`, the Whittaker polynomial whose s-derivative produces the logarithms.
- Numerical parameters matter. The horocycle height must satisfy $\Im\tau \geq 1$, or some coset is carried into the pole at ∞ and the terms overwhelm their sum. The number of sample points must beat the positive Fourier coefficients, which grow like $\exp(4\pi\sqrt{pm})$ for pole order p: on the pole-order-30 forms of $X_0^{15}(2)$ the principal-part gate reads $5.8 \cdot 10^{73}$, $1.2 \cdot 10^{62}$, $2.7 \cdot 10^{28}$ and $8.7 \cdot 10^{-103}$ at $K = 48$, 64, 96 and 192 respectively. Raising the working precision without raising K has no effect.
- Every reported value is gated on the principal-part check; a run failing that gate is discarded.
- The exact constant-term evaluation of §8 (`cusp1.m`, `cusp2.m`, and the two-channel solver `ledger2.py`) and the complete campaign data — principal-part dumps, ground-truth sweeps, and the constraints ledger — are banked on the branch `m0-theta-campaign` of the accompanying repository.

- The campaign of §9 is banked on the same branch under `vvdata/weyl-campaign`: the residue-pairing solver (`eis32.m`), the exact rationalisers (`kernelrat.py`, `eisrat.py`), the genus-average drivers (`allgross.m`, `genallgross.py`), the dictionary and mass probes (`grossid100.m`, `grossid900.m`, `swapprobe.m`, `massprobe.m`, `genusfit15sw.m`), the per-base evidence records (`mono/epool/eis32e/kernelrat/eisrat/allgross` per base), and the composite-s preregistration (`thetasum-form.md`, `mass210.log`).
- The bases with $M \geq 660$ are reachable only through `eis32k.m`, which replaces the numerical Weil-representation loop of `eis32.m` by the closed form of [7, Thm. 6.4] (agreement with the numerical driver checked end-to-end at $X_0^{210}(1)$: 960 of 960 words, identical solve residual $1.0087 \cdot 10^{-42}$). Its `EST` switch selects the tracked isotropic coset and is what produces the $\eta = 0$ rows of Remark 9.1; the corresponding control runs reproduce the banked η^* fits exactly. Its `CTL` switch recomputes every ρ-entry by the numerical route and reports the worst deviation; this is the gate through which the odd-level 2-part of Observation 9.8 was admitted (worst $1.3 \cdot 10^{-79}$ over all 144 words of $X_0^{15}(1)$, against $7.3 \cdot 10^{-80}$ for the unchanged even-level path at $X_0^{15}(2)$), run before any fit. Restricted-support gauge checks (the rank exclusions and the degeneracy quoted above) are generated by `genrestrict.py`.

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